Requirement Analysis Phase

Project Title: ComicCraft – AI Comic Story Creator Using Gemini Models

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1. Introduction

Requirement Analysis is the phase in which the needs, functions, resources, and expectations of the ComicCraft system are identified and documented. The main purpose is to clearly define what the system should do and what users require from it.

2. Project Requirements

ComicCraft should provide an easy-to-use platform that allows users to enter a story idea or prompt and receive AI-generated comic content using Gemini models.

3. Functional Requirements

- User

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3. Functional Requirements

- User registration/login if authentication is included.
- Accept story ideas, themes, characters, or prompts from the user.
- Generate comic storylines using Gemini models.
- Generate character descriptions and dialogues.
- Generate scene and panel descriptions.
- Allow users to edit generated content.
- Allow users to regenerate selected content.
- Display the generated comic story in an organized format.
- Provide save/export functionality where implemented.

4. Non-Functional Requirements

- User-friendly and simple interface.
- Fast and responsive system.
- Reliable communication with the Gemini service.
- Secure handling of user information and API credentials.
- Maintainable and scalable software design.
- Consistent and understandable generated content.

5. Hardware Requirements

- Computer or laptop for development and testing.
- Minimum 4 GB RAM; 8 GB or more is recommended.
- Internet connection for accessing cloud-based AI services.
- Sufficient storage for the application and project files.

6. Software Requirements

- Operating system such as Windows, Linux, or macOS.
- Programming language and framework selected for application development.
- Gemini API / Google AI services for AI-based generation.
- Code editor or IDE such as Visual Studio Code.

- Web browser for accessing and testing the application.
- Database or cloud storage if required by the implementation.

7. User Requirements

Users should be able to enter a simple idea and receive meaningful comic content without needing advanced writing or programming skills. The interface should make it easy to view, edit, and regenerate the generated content.

8. System Constraints

- AI output depends on the quality and clarity of the user prompt.
- Internet connectivity may be required for Gemini-based generation.
- API usage may be subject to service limits and availability.
- AI-generated content may require user review and editing.

9. Expected Outcome

The requirement analysis establishes a clear set of functional, non-functional, hardware, software, and user requirements for ComicCraft. These requirements provide the foundation for the subsequent project design and development phases.